

A Practical Guide to Logistic regression

TOPIC -- LOGISTIC REGRESSION

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$P(y = 1 | x) = \sigma(w^T x + b)$$

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Each of the probabilities except $p_0(x)$ will have their own set of regression coefficients β_n . It can be seen that, as required, the sum of the $p_n(x)$ over all categories n is 1. The selection of $p_0(x)$ to be defined in terms of the other probabilities is artificial. Any of the probabilities could have been selected to be so defined. This special value of n is termed the "pivot index", and the log-odds (tn) are expressed in terms of the pivot probability and are again expressed as a linear combination of the explanatory variables:

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$$\begin{aligned} \Pr(Y_i = 1) &= \frac{e^{(\beta_1 + C) \cdot X_i}}{e^{(\beta_1 + C) \cdot X_i} + e^{(\beta_0 + C) \cdot X_i}} = \frac{e^{\beta_1 \cdot X_i} \cdot e^{C \cdot X_i}}{e^{\beta_1 \cdot X_i} \cdot e^{C \cdot X_i} + e^{\beta_0 \cdot X_i} \cdot e^{C \cdot X_i}} = \frac{e^{\beta_1 \cdot X_i}}{e^{\beta_1 \cdot X_i} + e^{\beta_0 \cdot X_i}} \\ \Pr(Y_i = 1) &= \frac{e^{\beta_1 \cdot X_i} \cdot e^{C \cdot X_i}}{e^{\beta_1 \cdot X_i} \cdot e^{C \cdot X_i} + e^{\beta_0 \cdot X_i} \cdot e^{C \cdot X_i}} = \frac{e^{\beta_1 \cdot X_i}}{e^{\beta_1 \cdot X_i} + e^{\beta_0 \cdot X_i}} \\ \Pr(Y_i = 1) &= \frac{e^{\beta_1 \cdot X_i} \cdot e^{C \cdot X_i}}{e^{\beta_1 \cdot X_i} \cdot e^{C \cdot X_i} + e^{\beta_0 \cdot X_i} \cdot e^{C \cdot X_i}} = \frac{e^{\beta_1 \cdot X_i}}{e^{\beta_1 \cdot X_i} + e^{\beta_0 \cdot X_i}} \end{aligned}$$

$$\}_{1}\cdot \mathbf{X}_{i}}\}.\end{aligned}}}$$

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and the maximization procedure can be accomplished by solving the above two equations for β_0 and β_1 , which, again, will generally require the use of numerical methods. The values of β_0 and β_1 which maximize L using the above data are found to be:

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In a Bayesian statistics context, prior distributions are normally placed on the regression coefficients, for example in the form of Gaussian distributions. There is no conjugate prior of the likelihood function in logistic regression. When Bayesian inference was performed analytically, this made the posterior distribution difficult to calculate except in very low dimensions. Now, though, automatic software such as OpenBUGS, JAGS, PyMC, Stan or Turing.jl allows these posteriors to be computed using simulation, so lack of conjugacy is not a concern. However, when the sample size or the number of parameters is large, full Bayesian simulation can be slow, and people often use approximate methods such as variational Bayesian methods and expectation propagation.

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Generalizations This simple model is an example of binary logistic regression, and has one explanatory variable and a binary categorical variable which can assume one of two categorical values. Multinomial logistic regression is the generalization of binary logistic regression to include any number of explanatory variables and any number of categories.

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$$\text{logit}(\mathbb{E}[Y_i | x_{1,i}, \dots, x_{m,i}]) = \text{logit}(p_i) = \ln\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_m x_{m,i}$$

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Deviance and likelihood ratio test a simple case In any fitting procedure, the addition of another fitting parameter to a model (e.g. the beta parameters in a logistic regression model) will almost always improve the ability of the model to predict the measured outcomes. This will be true even if the additional term has

no predictive value, since the model will simply be "overfitting" to the noise in the data. The question arises as to whether the improvement gained by the addition of another fitting parameter is significant enough to recommend the inclusion of the additional term, or whether the improvement is simply that which may be expected from overfitting. In short, for logistic regression, a statistic known as the deviance is defined which is a measure of the error between the logistic model fit and the outcome data. In the limit of a large number of data points, the deviance is chi-squared distributed, which allows a chi-squared test to be implemented in order to determine the significance of the explanatory variables. Linear regression and logistic regression have many similarities. For example, in simple linear regression, a set of K data points (x_k, y_k) are fitted to a proposed model function of the form $y = b_0 + b_1 x$. The fit is obtained by choosing the b parameters which minimize the sum of the squares of the residuals (the squared error term) for each data point:

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Deviance and likelihood ratio tests In linear regression analysis, one is concerned with partitioning variance via the sum of squares calculations – variance in the criterion is essentially divided into variance accounted for by the predictors and residual variance. In logistic regression analysis, deviance is used in lieu of a sum of squares calculations. Deviance is analogous to the sum of squares calculations in linear regression and is a measure of the lack of fit to the data in a logistic regression model. When a "saturated" model is available (a model with a theoretically perfect fit), deviance is calculated by comparing a given model with the saturated model. This computation gives the likelihood-ratio test: